CAR4 – ROS2 to Raspberry Pi Motor Control Setup

1. System Architecture

```
Laptop / Ubuntu / ROS 2 Jazzy
        |
        | ROS 2 topic: /car4/cmd
        |
        v
ROS2 → Raspberry Pi Bridge
        |
        | TCP :5000
        | Wi-Fi
        v
Raspberry Pi
192.168.0.51
        |
        | /dev/ttyUSB0
        | USB Serial 115200
        v
Arduino Nano
        |
        v
TB6612FNG Motor Driver
        |
        v
CAR4 Motors
```

2. Raspberry Pi Information

Raspberry Pi IP address(It will be different for each one)

```
192.168.0.51
```

Check it with:

```
hostname –I
```

Arduino serial port

```
/dev/ttyUSB0
```

Check it with:

```
ls -l /dev/ttyUSB*
```

Serial baud rate

115200

3. Verify Python

On the Raspberry Pi:

```
python3 -c "import sys; print(sys.version)"
```

Expected:

```
3.13.5
```

Check terminal-control support:

```
python3 -c "import termios, tty; print('terminal control OK')"
```

Expected:

```
terminal control OK
```

4. Test Arduino Directly From Raspberry Pi

Before using ROS2, verify that the Pi can control the Arduino.

```
python3 - <<'PY'
import serial
import time

ser = serial.Serial('/dev/ttyUSB0', 115200, timeout=1)
time.sleep(2)

print("Sending F,100...")
ser.write(b'F,100\n')

time.sleep(2)

print("Sending STOP...")
ser.write(b'STOP\n')

ser.close()

print("Done")
PY
```

If the car moves forward and stops, the Pi → Arduino connection is working.

5. Raspberry Pi Network Bridge

The Pi does **not** need ROS2 installed.

The Pi simply receives commands over Wi-Fi and forwards them to the Arduino.

Run this on the **Raspberry Pi SSH terminal**:

```
python3 - <<'PY'
import socket
import serial

HOST = "0.0.0.0"
```

```
PORT = 5000

ser = serial.Serial("/dev/ttyUSB0", 115200, timeout=1)

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
server.bind((HOST, PORT))
server.listen(1)

print(f"Bridge listening on port {PORT}...")
print("Waiting for laptop...")

while True:
    conn, addr = server.accept()
    print("Connected:", addr)

    try:
        while True:
            data = conn.recv(1024)

            if not data:
                break

            print("Received:", repr(data))

            ser.write(data)
            ser.flush()

    except Exception as e:
        print("Error:", e)

    finally:
        conn.close()
        print("Laptop disconnected")

PY
```

Keep this terminal running.

You should see:

```
Bridge listening on port 5000...
Waiting for laptop...
```

6. ROS2 Installation

ROS2 is installed on the **laptop**, not the Raspberry Pi.

On the laptop Ubuntu terminal:

```
source /opt/ros/jazzy/setup.bash
```

Check ROS2:

```
ros2 topic list
```

You should see at least:

```
/parameter_events
/rosout
```

7. ROS2 → Raspberry Pi Bridge

Create the bridge directory:

```
mkdir -p ~/car4_bridge
```

Create the Python file:

```
nano ~/car4_bridge/ros_pi_bridge.py
```

Put this code inside:

```

import socket

import rclpy from rclpy.node import Node from std_msgs.msg import
String

PI_IP = "192.168.0.51" PI_PORT = 5000

class ROSPiBridge(Node):

def __init__(self):
    super().__init__('ros_pi_bridge')

    self.subscription = self.create_subscription(
        String,
        '/car4/cmd',
        self.command_callback,
        10
    )
```

```
    self.sock = None

    self.connect_to_pi()

def connect_to_pi(self):

    try:
        self.sock = socket.create_connection(
            (PI_IP, PI_PORT),
            timeout=5
        )

        self.get_logger().info(
            f'Connected to Pi at {PI_IP}:{PI_PORT}'
        )

    except Exception as e:

        self.get_logger().error(
            f'Could not connect to Pi: {e}'
        )

def command_callback(self, msg):

    command = msg.data.strip()

    self.get_logger().info(
        f'Sending command: {command}'
    )

    try:

        if self.sock is None:
            self.connect_to_pi()

        if self.sock:

            self.sock.sendall(
                (command + '\n').encode()
            )

    except Exception as e:

        self.get_logger().error(
            f'Failed to send command: {e}'
```

```
        )

        self.sock = None


def main(args=None):

rclpy.init(args=args)

node = ROSPiBridge()

try:

    rclpy.spin(node)

except KeyboardInterrupt:

    pass

if node.sock:
    node.sock.close()

node.destroy_node()

rclpy.shutdown()


if name == 'main': main()
```

Save:

```
Ctrl + O
Enter
Ctrl + X
```

8. Start the ROS2 Bridge

On the **laptop Ubuntu terminal**:

```
source /opt/ros/jazzy/setup.bash
```

Then:

```
python3 ~/car4_bridge/ros_pi_bridge.py
```

Expected:

```
[INFO] [...] [ros_pi_bridge]: Connected to Pi at 192.168.0.51:5000
```

Keep this terminal running.

9. CAR4 Commands

Open a **second Ubuntu terminal** on the laptop.

First:

```
source /opt/ros/jazzy/setup.bash
```

Now you can control CAR4 using ROS2.

Forward

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'F,100'}"
```

Backward

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'B,100'}"
```

Left

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'L,100'}"
```

Right

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'R,100'}"
```

Stop

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'STOP'}"
```

10. Command Reference

Command	Function
F,100	Forward at speed 100
B,100	Backward at speed 100
R,100	Right at speed 100
L,100	Left at speed 100
STOP	Stop motors

The speed can be changed.

For example:

For example:

ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'F,50'}"

Forward at speed 50.

ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'F,150'}"

Forward at speed 150.

ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'F,200'}"

Forward at speed 200.

11. Complete Startup Procedure

Every time you want to operate CAR4:

Terminal 1 — Raspberry Pi SSH

Start the Pi bridge:

```
python3 - <<'PY'
import socket
import serial

HOST = "0.0.0.0"
PORT = 5000

ser = serial.Serial("/dev/ttyUSB0", 115200, timeout=1)

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
server.bind((HOST, PORT))
server.listen(1)

print(f"Bridge listening on port {PORT}...")
print("Waiting for laptop...")

while True:
    conn, addr = server.accept()
    print("Connected:", addr)
```

```
    try:
        while True:
            data = conn.recv(1024)

            if not data:
                break

            print("Received:", repr(data))

            ser.write(data)
            ser.flush()

    except Exception as e:
        print("Error:", e)

    finally:
        conn.close()
        print("Laptop disconnected")
PY
```

Terminal 2 — Laptop Ubuntu

Start ROS2 bridge:

```
source /opt/ros/jazzy/setup.bash
python3 ~/car4_bridge/ros_pi_bridge.py
```

Terminal 3 — Laptop Ubuntu

Control CAR4:

```
source /opt/ros/jazzy/setup.bash
```

Then:

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'F,100'}"
```

or:

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'B,100'}"
```

or:

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'L,100'}"
```

or:

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'R,100'}"
```

and always use:

```
ros2 topic pub --once /car4/cmd std_msgs/msg/String "{data: 'STOP'}"
```

to stop the car.

12. Important Notes

- **ROS2 runs on the laptop.**
- **The Raspberry Pi does not currently have ROS2 installed.**
- Pi IP: 192.168.0.51
- Pi serial: /dev/ttyUSB0
- Serial baud: 115200
- TCP port: 5000
- ROS2 topic: /car4/cmd
- Message type: std_msgs/msg/String

The current setup has been successfully tested end-to-end with F,100.